

```

sam@sam-VirtualBox:~$ sudo systemctl enable named.service
Synchronizing state of named.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable named
sam@sam-VirtualBox:~$ sudo systemctl start named.service
sam@sam-VirtualBox:~$ sudo systemctl status named.service
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-12-13 19:08:43 EST; 13min ago
     Docs: man:named(8)
    Main PID: 4340 (named)
      Status: "running"
       Tasks: 5 (limit: 2264)
      Memory: 23.4M (peak: 23.6M)
         CPU: 44ms
    CGroup: /system.slice/named.service
            └─4340 /usr/sbin/named -f -u bind

déc. 13 19:08:43 sam-VirtualBox named[4340]: managed-keys-zone: loaded serial 0
déc. 13 19:08:43 sam-VirtualBox named[4340]: zone 127.in-addr.arpa/IN: loaded serial 1
déc. 13 19:08:43 sam-VirtualBox named[4340]: zone 0.in-addr.arpa/IN: loaded serial 1
déc. 13 19:08:43 sam-VirtualBox named[4340]: zone 255.in-addr.arpa/IN: loaded serial 1
déc. 13 19:08:43 sam-VirtualBox named[4340]: zone localhost/IN: loaded serial 2
déc. 13 19:08:43 sam-VirtualBox named[4340]: all zones loaded
déc. 13 19:08:43 sam-VirtualBox systemd[1]: Started named.service - BIND Domain Name Server.
déc. 13 19:08:43 sam-VirtualBox named[4340]: running
déc. 13 19:08:43 sam-VirtualBox named[4340]: managed-keys-zone: Initializing automatic trust anchor management for zone-
déc. 13 19:08:43 sam-VirtualBox named[4340]: managed-keys-zone: Initializing automatic trust anchor management for zone-
lines 1-22/22 (END)
[1]~ Arrêté                  sudo systemctl status named.service

```

- Configuration de la zone dns

```
GNU nano 7.2 /etc/bind/db.Gabte
;
; BIND data file for local loopback interface
;
$TTL      604800
@         IN      SOA      ns.Gabtech.local. admin.Gabtech.local. (
                                2          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ns.Gabtech.local.
@         IN      A        192.168.1.200
@         IN      A        192.168.1.200
;
pc1       IN      A        192.168.1.20
pc2       IN      A        192.168.1.21
pc3       IN      A        192.168.1.22
pc4       IN      A        192.168.1.23
pc5       IN      A        192.168.1.24
web       IN      A        192.168.1.25
dns       IN      A        192.168.1.26
mail      IN      A        192.168.1.27
;
www       IN      CNAME    web.Gabtech.local.
```

- Configuration de la zone inverse

```
GNU nano 7.2 /etc/bind/db.19
;
; BIND reverse data file for local loopback interface
;
$TTL      604800
@         IN      SOA      ns.Gabtech.local. admin.lab.local. (
                                1          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ns.Gabtech.local.

200       IN      PTR      ns.Gabtech.local.
20        IN      PTR      pc1.Gabtech.local.
21        IN      PTR      pc2.Gabtech.local.
22        IN      PTR      pc3.Gabtech.local.
23        IN      PTR      pc4.Gabtech.local.
24        IN      PTR      pc5.Gabtech.local.
25        IN      PTR      web.Gabtech.local.
26        IN      PTR      dns.Gabtech.local.
27        IN      PTR      mail.Gabtech.local.
```

- Tests de configuration

```

sam@sam-VirtualBox:~$ sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/db.192.168.1
zone 1.168.192.in-addr.arpa/IN: loaded serial 1
OK
sam@sam-VirtualBox:~$ sudo nano /etc/bind/db.Gabtech.local
sam@sam-VirtualBox:~$ sudo named-checkzone Gabtech.local /etc/bind/db.Gabtech.local
zone Gabtech.local/IN: loaded serial 2
OK
sam@sam-VirtualBox:~$ sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/db.192.168.1
zone 1.168.192.in-addr.arpa/IN: loaded serial 1
OK

```

- Test de redémarrage

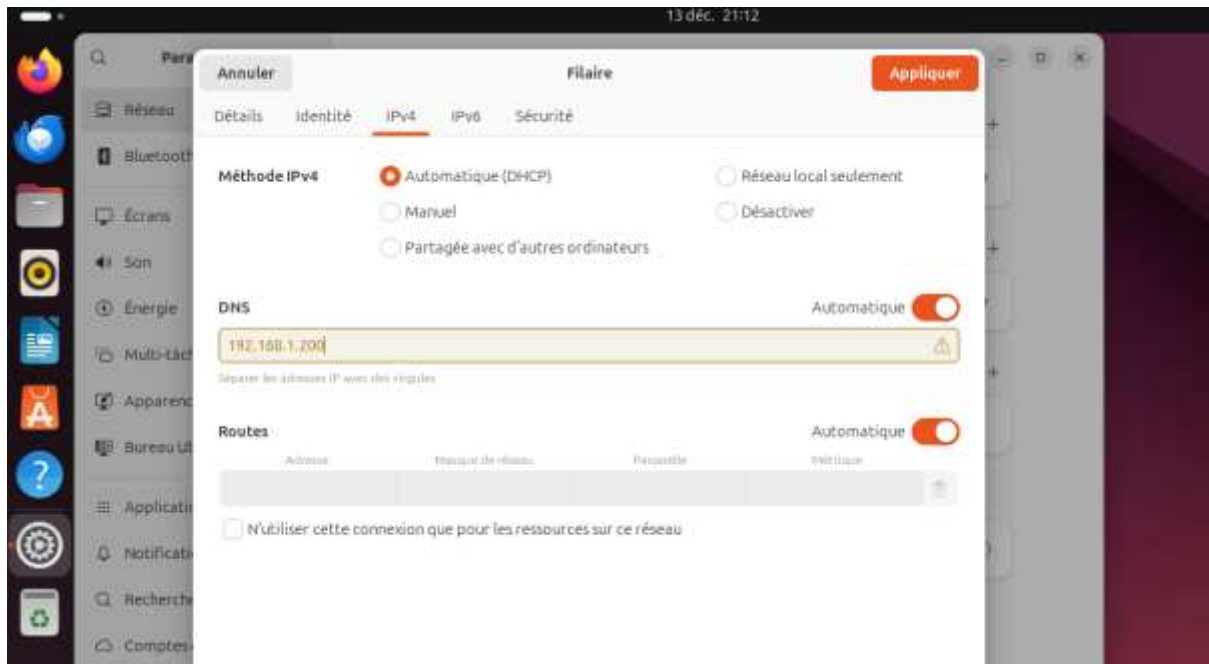
```

sam@sam-VirtualBox:~$ sudo systemctl restart bind9
sam@sam-VirtualBox:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-12-13 21:00:53 EST; 11s ago
     Docs: man:named(8)
  Main PID: 5520 (named)
    Status: "running"
     Tasks: 4 (limit: 2264)
  Memory: 22.4M (peak: 22.7M)
     CPU: 62ms
    CGroup: /system.slice/named.service
            └─5520 /usr/sbin/named -f -u bind

déc. 13 21:00:53 sam-VirtualBox named[5520]: zone 255.in-addr.arpa/IN: loaded serial 1
déc. 13 21:00:53 sam-VirtualBox named[5520]: zone 0.in-addr.arpa/IN: loaded serial 1
déc. 13 21:00:53 sam-VirtualBox named[5520]: zone localhost/IN: loaded serial 2
déc. 13 21:00:53 sam-VirtualBox named[5520]: zone 127.in-addr.arpa/IN: loaded serial 1
déc. 13 21:00:53 sam-VirtualBox named[5520]: zone 1.168.192.in-addr.arpa/IN: loaded serial 1
déc. 13 21:00:53 sam-VirtualBox named[5520]: all zones loaded
déc. 13 21:00:53 sam-VirtualBox systemd[1]: Started named.service - BIND Domain Name Server.
déc. 13 21:00:53 sam-VirtualBox named[5520]: running
déc. 13 21:01:03 sam-VirtualBox named[5520]: managed-keys-zone: Unable to fetch DNSKEY set '.': timed out
déc. 13 21:01:03 sam-VirtualBox named[5520]: resolver prining query complete: timed out
sam@sam-VirtualBox:~$

```

- Configuration du Client



```
sam@sam-VirtualBox:~$ sudo systemctl restart NetworkManager
[sudo] Mot de passe de sam :
```

Test de résolution

- Test de la zone directe

```
sam@sam-VirtualBox:~$ dig Gabtech.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> Gabtech.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 21157
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;Gabtech.local.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Dec 13 21:14:49 EST 2025
;; MSG SIZE rcvd: 42
```

```

sam@sam-VirtualBox:~$ dig pc1.Gabtech.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> pc1.Gabtech.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 55866
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;pc1.Gabtech.local.          IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Dec 13 21:15:13 EST 2025
;; MSG SIZE rcvd: 46

```

```

sam@sam-VirtualBox:~$ dig mail.Gabtech.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> mail.Gabtech.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 11172
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;mail.Gabtech.local.        IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Dec 13 21:15:44 EST 2025
;; MSG SIZE rcvd: 47

```

- Test de la zone inverse

```

sam@sam-VirtualBox:~$ dig -x 192.168.1.25

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> -x 192.168.1.25
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 24430
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;25.1.168.192.in-addr.arpa.      IN      PTR

;; Query time: 52 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sat Dec 13 21:16:21 EST 2025
;; MSG SIZE rcvd: 54

```

- **Installation du serveur DHCP**

```

sam@sam-VirtualBox:~$ sudo apt install isc-dhcp-server
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Le paquet suivant a été installé automatiquement et n'est plus nécessaire :
  libnlvm19
Veuillez utiliser « sudo apt autoremove » pour le supprimer.
Les paquets supplémentaires suivants seront installés :
  isc-dhcp-common
Paquets suggérés :
  isc-dhcp-server-ldap polycoreutils
Les NOUVEAUX paquets suivants seront installés :
  isc-dhcp-common isc-dhcp-server
0 nis à jour, 2 nouvellement installés, 0 à enlever et 58 non mis à jour.
Il est nécessaire de prendre 1 281 ko dans les archives.
Après cette opération, 4 281 ko d'espace disque supplémentaires seront utilisés.
Souhaitez-vous continuer ? [O/n] o
Réception de :1 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-server amd64 4.4.3-P1-4ubuntu2 [1 236 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-common amd64 4.4.3-P1-4ubuntu2 [45,8 kB]
1 281 ko réceptionnés en 11s (114 ko/s)
Préconfiguration des paquets...
Sélection du paquet isc-dhcp-server précédemment désélectionné.
(Lecture de la base de données... 289089 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ...
Dépaquetage de isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Sélection du paquet isc-dhcp-common précédemment désélectionné.
Préparation du dépaquetage de .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ...
Dépaquetage de isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Paramétrage de isc-dhcp-server (4.4.3-P1-4ubuntu2) :

```

- **Configuration, redémarrage et vérification**


```

sam@sam-VirtualBox: $ sudo nano /etc/default/isc-dhcp-server
sam@sam-VirtualBox: $ sudo nano /etc/dhcp/dhcpd.conf
sam@sam-VirtualBox: $ sudo systemctl restart isc-dhcp-server
sam@sam-VirtualBox: $ sudo systemctl status isc-dhcp-server
* isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Sat 2025-12-13 21:37:22 EST; 23s ago
   Duration: 47ms
   Docs: man:dhcpd(8)
   Process: 7029 ExecStart=/bin/sh -ec CONFIG_FILE=/etc/dhcp/dhcpd.conf; if [ -f /etc/ltsp/dhcpd.conf ]; the
   Main PID: 7029 (code=exited, status=1/FAILURE)
   CPU: 17ms

déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]:
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]: If you think you have received this message due to a bug rather
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]: than a configuration issue please read the section on submitting
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]: bugs on either our web page at www.isc.org or in the README file
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]: before submitting a bug. These pages explain the proper
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]: process and the information we find helpful for debugging.
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]:
déc. 13 21:37:22 sam-VirtualBox dhcpd[7029]: exiting.
déc. 13 21:37:22 sam-VirtualBox systemd[1]: isc-dhcp-server.service: Main process exited, code=exited, status=1/FAILURE
déc. 13 21:37:22 sam-VirtualBox systemd[1]: isc-dhcp-server.service: Failed with result 'exit-code'.
...skipping...
* isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Sat 2025-12-13 21:37:22 EST; 23s ago

```

- Analyse des logs

```

sam@sam-VirtualBox: $ sudo systemctl status isc-dhcp-server
sam@sam-VirtualBox: $ sudo dhcpd -t
Internet Systems Consortium DHCP Server 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
sam@sam-VirtualBox: $ journalctl -u isc-dhcp-server
déc. 13 21:25:47 sam-VirtualBox systemd[1]: Started isc-dhcp-server.service - ISC DHCP IPv4 server.
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: Internet Systems Consortium DHCP Server 4.4.3-P1
déc. 13 21:25:47 sam-VirtualBox sh[6761]: Internet Systems Consortium DHCP Server 4.4.3-P1
déc. 13 21:25:47 sam-VirtualBox sh[6761]: Copyright 2004-2022 Internet Systems Consortium.
déc. 13 21:25:47 sam-VirtualBox sh[6761]: All rights reserved.
déc. 13 21:25:47 sam-VirtualBox sh[6761]: For info, please visit https://www.isc.org/software/dhcp/
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: Copyright 2004-2022 Internet Systems Consortium.
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: All rights reserved.
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: For info, please visit https://www.isc.org/software/dhcp/
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: Config file: /etc/dhcp/dhcpd.conf
déc. 13 21:25:47 sam-VirtualBox sh[6761]: Config file: /etc/dhcp/dhcpd.conf
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: Database file: /var/lib/dhcp/dhcpd.leases
déc. 13 21:25:47 sam-VirtualBox sh[6761]: Database file: /var/lib/dhcp/dhcpd.leases
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: PID file: /run/dhcp-server/dhcpd.pid
déc. 13 21:25:47 sam-VirtualBox sh[6761]: PID file: /run/dhcp-server/dhcpd.pid
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]: Wrote 0 leases to leases file.
déc. 13 21:25:47 sam-VirtualBox sh[6761]: Wrote 0 leases to leases file.
déc. 13 21:25:47 sam-VirtualBox dhcpd[6761]:

```